



RoleVR: Multi-experience in immersive virtual reality between co-located HMD and non-HMD users

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Abstract

In this study, we present RoleVR, which can provide a similar high level of presence and multi experience for co-located head-mounted display (HMD) and Non-HMD users in an asymmetric virtual reality (VR) environment. The core of RoleVR is distinguishing the difference between the asymmetric environments (in terms of the system and the experience) of HMD and Non-HMD users to design optimized roles for these users. Here, we assign HMD user with spatial role that maximizes the sense of space based on three-dimensional visual information, and we assign Non-HMD user with temporal role in which they take control of communication and action, and understand the overall situation according to the flow of time. We also design an interaction for walking and a hand interface to enhance presence. This is achieved by understanding the user's role, thereby improving the immersion. Finally, we created an asymmetric VR application that considers the interaction between roles and performed survey-based experiments to verify the basic presence and multi-experience of users in RoleVR. Through this process, we confirmed that RoleVR provides satisfactory presence for co-located HMD and Non-HMD users, and a variety of experiences specialized for each role.

Keywords Immersive virtual reality · RoleVR · Multi-experience · Asymmetric virtual reality · Co-located users · Presence

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1 Introduction

Recent advances in virtual reality head-mounted display (HMD) technology (e.g. Oculus Rift and HTC Vive) have made it possible to provide users with more realistic experiences in virtual environments created by computers. Based on this, research is being conducted on immersive virtual reality (IVR) in which interaction occurs via virtual objects, environments and a variety of methods (eye gaze, hand, and walking) [11]. A key factor in user oriented immersive virtual reality (IVR) is presence. Here, presence is a phenomenon whereby users behave and feel as if they perceive the virtual reality environment as actual reality (i.e. “being there”). In order to experience a high level of presence, such that a user finds it difficult to distinguish between the real and virtual worlds, it is necessary to provide an environment that is more immersed in the virtual environment. Therefore, interaction techniques that enhance the immersion should be supported to provide an improved presence [31].

Immersion studies to improve presence in IVR have been developed for satisfying the users’ five senses. Along with research on satisfying the visual sense through 3D monitors, multi-projectors, and HMDs, which can transmit high-resolution images in 3D, there has also been research on applications that capture HMD users’ eye movements [29] and facial expression changes [22] to increase visual presence. Here, research is being conducted on satisfying the auditory sense in large-area IVR using the area and volumetric sources [28], and research is also being conducted on haptic systems that capture hand motions in detail to recreate them in virtual space and provide feedback through the physical responses that occur during interactions via hand [40]. In addition, motion-based IVR researches [20, 35] are developing the walking simulators or algorithms that allow the user to move freely in a large virtual reality space rather than a limited space.

It should be noted here is that, for most IVRs, the environment, system, interactions, etc. are built to focus on a single user’s five senses. Because of this, VR applications and IVR research (haptic devices, motion platforms) start with the user wearing the HMD of 1st-person view. However, the level or role of participation in VR can vary according to the user (some want to watch, some want to have certain forms of interaction, some want to be part of the experience), so VR systems may need to include a broad range of types of participation [38]. To do this, the focus of IVR research must be expanded beyond the five senses to include communication and roles. In this process, Non-HMD user can participate in VR depending on their role or situations. Research on collaborative virtual environments (CVE) proposes interactions through which multiple users can collaborate in the same conditions and environment [5]. Furthermore, this research is developing in directions that present different experience environments, such as research [12] that divides up roles and proposes interactions between HMD users in VR and PC users who use a traditional GUI. ShareVR, which was recently proposed by Gugenheimer et al. [9] makes it possible for HMD user and Non-HMD users to have co-located asymmetric interaction. It was confirmed that all users felt a satisfying experience in the same virtual environment. However, research on asymmetric VR in CVE is still centered on HMD users, and Non-HMD users play an auxiliary role closer to designers than participants, or the Non-HMD user’s experience environment is limited in that it is non-immersive, so there is inevitably a large difference in the two users’ presence. As a result, the degree to which Non-HMD users can share in the VR experience is limited. However, studies [6] have been conducted steadily to create an experience environment that enhances the presence of Non-HMD users in nonimmersive environments through displays, such as 3D monitors, and multiproject, or cave automatic virtual environment (CAVE) from a viewpoint different from that of HMD users who are provided with a high spatial feeling through stereoscopic visual information from the HMD using

a binocular camera. In this respect, the Maze Commander proposed by Sajjadi et al [27] also demonstrated that everyone can be satisfied and immersed in the virtual environment if Non-HMD users are given independent roles in an environment where Non-HMD users can play together with HMD users. Therefore, in this study, it is suggested that, if HMD and Non-HMD users can share their experiences through independent roles and immersive interactions supporting them, the presence of Non-HMD users can be improved, as well as the presence of HMD users. Therefore, this study focuses on the presence of “being there together” for Non-HMD users beyond the presence of HMD users (“being there”).

This study recognizes the difference in the environment experienced by co-located HMD and Non-HMD users, and assigns different roles to such users, in addition to providing an immersive environment suitable for each situation. Therefore, we propose RoleVR. In the proposed system, users can experience similar experiences in the same virtual reality environment and multi-experiences in the process of immersing together. The core of this is allowing two users (HMD and Non-HMD) who are experiencing one VR application to both feel a satisfactory presence, enjoyment, and social interaction while having a variety of different experiences. For this to happen, we defined a role as an important factor in VR. Furthermore, we designed an interaction that is appropriate for the role (Fig. 1). The proposed RoleVR makes the following contributions.

1. **Role of RoleVR:** The behaviors associated with thoughts, situations, and decisions are shared so that all users can feel a similar presence regardless of whether they are wearing the HMD or not. Based on this, HMD user, who has 3D information, perceive the current situation and are assigned the task of movement, which is a spatial role, and Non-HMD user, who exists in a relatively non-immersive environment, perceive the overall situation and are in charge of an action, which is a temporal role.
2. **Immersive Interaction of RoleVR:** Interactions which can be more immersive in situations where all users are suitable for the role are designed. HMD user uses a walking simulator and is provided with the experience of actual movement. Non-HMD user is made to feel a high degree of presence while carrying out her/his actions which include overall situation control through hand interfaces.
3. **Analysis of RoleVR:** To verify the proposed RoleVR, we first created an asymmetric VR application. We also checked to see if the two users felt a similar level of presence in

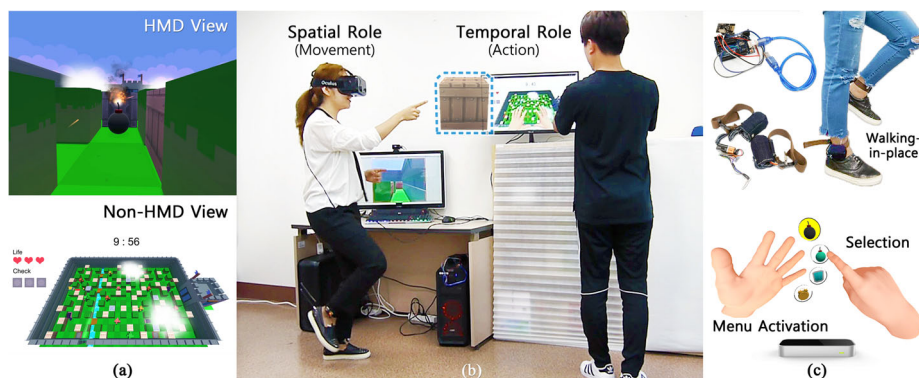


Fig. 1 RoleVR enables multi-experience with a similar presence for users wearing HMD and users without HMD: **a** asymmetric VR application production, **b** role based experience environment using communication and interaction, and **c** immersive interaction suitable for role of HMD or Non-HMD user

the same virtual environment. Then, we checked if each user felt an experience appropriate for their role, and we checked what kind of effect the roles and interactions had during the process. Here, the verification process was done through survey experiments on general participants.

2 Related work

2.1 Immersive virtual reality

IVR has been studied with the aim of presenting a virtual environment in which the user's experience of where they are, who they are with, and what they are doing is realized through the five senses. A variety of studies on increasing realism, including the displays that can transmit the virtual environment's visual information to the user in 3D and with presence, auditory sense using spatial audio, tactile output using haptic feedback, and movement through motion platforms, have been conducted [3, 28].

As advanced HMDs such as Oculus Rift and low price mobile platform HMDs such as GearVR have become popularized, it is no longer difficult for users to experience 3D visual information in a virtual environment. As such, current IVR provides users with basic presence. However, in addition to visual information and immersive techniques that realistically depict a virtual environment, interaction with objects and movement in virtual space are also necessary to provide a high presence in virtual reality. In order to design realistic interactions between the virtual and the real, the movement of body joints must be accurately detected and the intention of actions must be analyzed. In their research, Zhao et al. [40] used optical markers and Kinect cameras to capture the movement of actual hands and fingers, and reflect it in a virtual environment. In their research on movement, Vasylevska et al. [35] proposed a flexible space which provides infinite walking in a limited space. Lee et al. [20] created a walking simulator based on walking-in-place in which the user can freely move over a large VR terrain in a limited space. This kind of research makes it possible for users to have a variety of interactions with virtual objects and environments. The technique for providing feedback in terms of a physical response such as force or tactile sensations is an important part of IVR research as well. Carvalho et al. [3] performed a research on an interaction system which maps an actual hand to a virtual hand so that the experience of controlling virtual objects is similar to reality. However, most of the research requires additional VR systems or environments beyond HMD, so there are limits on their use as applications. Nevertheless, there is a need for immersive interactions involving the user walking or the user's hand interactions for the purpose of presence in IVR. RoleVR allows us to design simply structured walking and hand interactions to maximize the immersion of co-located users in limited space.

2.2 Collaborative virtual environments

CVE are distributed VR systems which allow the user to interact between the environments or between one user and another, and they were first defined by Churchill et al. [5]. Initially there were several research efforts, including Carlsson et al. [2], which focused mainly on distributed systems and were related to interactions between multiple users, such as distributed interactive virtual environments [26]. These distributed environments developed into asymmetric CVE in which the roles of several users are differentiated. For example, Johansen et al. [15] proposed a CVE which was focused on co-located synchronous interaction.

Furthermore, various application studies focused on collaboration have been conducted, such as HeatMeUp, which provides efficient pathfinding in a virtual environment through collaborations in which partners provide waypoint signals [24], and collaborative navigation, which enables disaster relief personnel to make rapid decisions in dangerous environments [1]. Later co-located CVE research has begun making discoveries which have increased the range and intention of activities which can be performed in a virtual environment by several users simultaneously [9]. Within this process, asymmetry through users' roles and whether they are equipped with HMD have become an important aspect of CVE, as the aim of RoleVR.

Duval et al. [7] presented asymmetric 2D/3D interactions so that HMD users and PC users could interact, and Oda et al. [25] proposed asymmetric interactions between local users equipped with AR HMD and remote users. This asymmetric CVE research differentiates the interactions and experiences between Non-HMD users and HMD users. In the Dollhouse VR proposed by Ibayashi et al. [12], Non-HMD users are designers who create the virtual space, and HMD users move around in the created virtual space and experience it. In this process, the both types of users are made to share information about the space.

The TurkDeck [4] study proposed by Cheng et al. uses slightly different approaches, in that both aim to allow human actuators to create physical/haptic feedback for HMD users. Human actuators exist in a space, like the HMD user, but they are only used as props for providing a highly immersive experience to the HMD user. Moreover, studies to improve the participation of Non-HMD users have been conducted, such as FaceDisplay [10], which is a multi-display mobile VR HMD that enables Non-HMD users to recognize the virtual environment of HMD users and interact with them. Gugenheimer et al. [9] proposed ShareVR, which combines an asymmetric experience with haptic feedback roles for Non-HMD users. This enables co-located asymmetric interaction between users equipped with HMD and users not equipped with HMD, and through this, co-located experiences were provided for the two users. Like ShareVR, RoleVR also has the feature of allowing co-located asymmetric interactions between HMD user and Non-HMD users. Co-located VR games that provide asymmetric experiences, such as Black Hat Cooperative and Ruckus Ridge VR Party [8, 34], have been released, and Sajjadi et al. [27] have presented a collaborative asymmetric game called *Maze Commander* that uses the Sifteo Cube. However, CVE research on asymmetric VR games and asymmetric interactions is limited, in that it provides non-immersive experience environments for Non-HMD users. Therefore, it has limitations in its ability to have HMD user and Non-HMD user be immersed in a VR perspective together, and to feel the same level of presence. ShareVR [9] is an extension of the work proposed by Cheng et al. [4] and gives Non-HMD users a broad range of activities, but as the Non-HMD roles do not go beyond auxiliary roles (designer concept), it does not overcome the difference in presence experienced by Non-HMD and HMD users. For the proposed RoleVR system, we aimed to design a system that overcomes these limitations so that Non-HMD user can feel a high level of presence in the CVE while at the same time they are given clear roles and immersive interactions for those roles, which allow them to feel a multi-experience. In addition, we aim to present a new direction for human-centered VR applications similar to the approach that Violante et al. [37] intended to implement by broadening the user base and providing various roles.

2.3 Presence

Presence is the most important factor a user must experience in VR. Slater et al. [32] in particular have conducted a wide range of studies on presence in IVR from a variety of

perspectives. They have compared presence by comparing virtual navigation using neural networks for using walking in IVR and optical tracker equipment [32], and analyzed the effect of communication through eye gaze on immersion and presence in VR [36]. Those studies have led to other studies on the differences in thoughts and behaviors of users according to the environment and conditions they encounter in VR [30], as well as studies analyzing the presence that users can experience in VR through viewpoint, muscle activity, etc [18]. Recently, the effect of VR throughout our lifestyles has been analyzed using a variety of approaches, such as psychology, and sociology [31]. Recently, studies have been conducted from various perspectives, such as analyzing the experiences at the moment when the VR ends and is changed to the real world [17].

Based on these studies, researchers have recently studied applications that analyze the immersion of a user's interactions such as hand interactions and walking as well as presence in various situations in virtual environments [20]. For RoleVR, we designed an experiment process that can systematically analyze the presence and multi-experience between HMD and Non-HMD users in IVR based on existing presence research that showed differences can occur owing to the environment, situation, or roles.

3 Asymmetric VR application

3.1 Overview

The purpose of RoleVR is to have the co-located HMD and Non-HMD users feel a similar level of presence and to provide an optimal experience to the users. Toward this end, we need an asymmetric VR application which can verify RoleVR's purposes. If we look at existing asymmetric co-located VR games, the relationship between HMD and Non-HMD users can be classified into conflictive and cooperative. Ruckus Ridge VR Party [8] is an asymmetric VR application with a typical conflictive structure in which HMD and Non-HMD users must both hinder their counterparts' activities to achieve their own goals. Here, the Non-HMD user's experience environment is not very different from operations in the existing 3D games, so there may be a partial improvement in enjoyment due to the active participation of HMD user; however, from a VR perspective, it is difficult to expect an improvement in immersion. On the other hand, in cooperative applications with shared goals, such as Black Hat Cooperative [34], the HMD and Non-HMD users share all information (including 1st person point of view videos) so even Non-HMD user in a non-immersive environment experience part of the presence of VR. RoleVR is based on existing applications and was designed as a cooperative application to focus more on the immersion of Non-HMD user in IVR.

We decided on a structure for communication between HMD and Non-HMD users through viewpoint and speech, which is needed to create a concrete asymmetric co-located application. In an online survey by Gugenheimer et al. [9], speech was recognized as the most important factor in asymmetric interaction between HMD and Non-HMD users. Therefore, all users are not equipped with headsets so that the process of application running can be shared through conversation. It was also shown that, from a gameplay perspective, having a different view of the virtual scene is important, so HMD user is provided with the 1st person view as with existing games, and Non-HMD user is provided with a different 3rd person view (top or perspective) from which they can understand the overall situation. However, in order to increase the Non-HMD user's presence, she/he can share the 1st

person scene of HMD user. Finally, we created active content to maximize use of the space in order to improve the overall presence of co-located users like those in ShareVR [9].

3.2 Design

The concept of the RoleVR application consists of a maze scene. Sajjadi et al. [27] and Jeong et al. [14] confirmed that mazes provide a new kind of presence in IVR. The survey used by Jeong et al. [14] confirmed that events appropriate for a maze can be used as an important factor in increasing the user's interest and immersion. Therefore, we combined communication between users with activities that utilize space via mazes and events to create a new asymmetric VR application that provides a high level of presence. Finally, the goals presented to the user in the application are to avoid NPCs and to quickly escape the maze by finding all the hidden check points. Here, the user uses bombs to perform tasks such as collecting items, removing NPCs, and creating maze paths.

Figure 2 shows the proposed RoleVR application, and several game factors are set in the maze pattern scene to be appropriate for asymmetric users. Basically, the maze is composed of walls and trees, which are fixed objects that cannot be removed, check points, which open the destination door, silver box objects, which conceal items, and walls, which can be removed with bombs.

4 RoleVR

RoleVR focuses on the presence and experience between co-located HMD and Non-HMD users. First, it is important that all users feel a similar level of presence in IVR. To make that happen, we must consider the concentration and immersion of Non-HMD users and not just HMD users. Therefore, in this study, we have defined roles and immersive interactions. The interactions are designed to be suitable for the roles to satisfy all users' presence together. Through this process, we also allow each user to have a variety of experiences in each situation (Fig. 3).

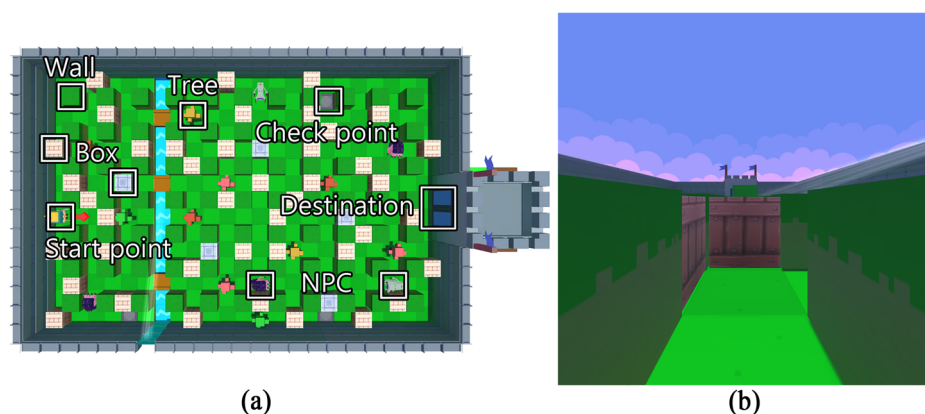


Fig. 2 The proposed RoleVR application: **a** construction of application consisted of various objects, **b** HMD user's viewpoint

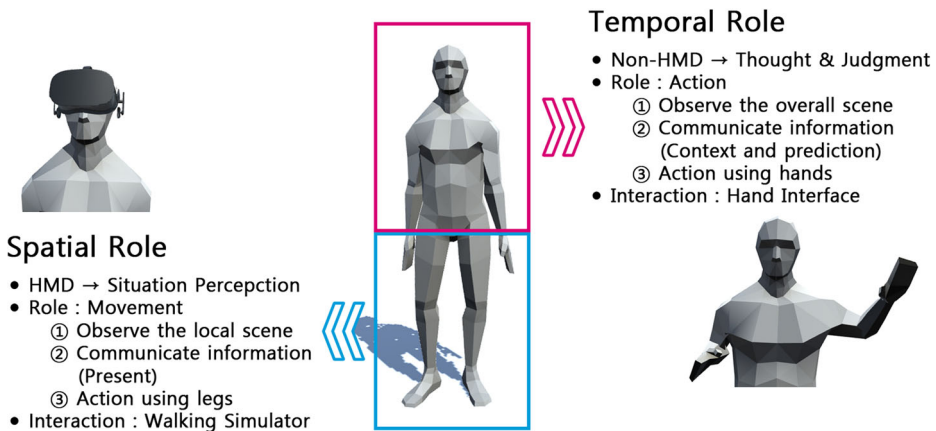


Fig. 3 The core structure of the proposed RoleVR with HMD and Non-HMD users

4.1 Role

To provide a similar level of presence between HMD and Non-HMD users, RoleVR first defines the roles. HMD user sees the virtual scenes from the 1st-person view so they feel a strong awareness of their current site and sense of space. On the other hand, Non-HMD user understands the overall situation through the 3rd person view, and the god-like interactions presented by Stafford et al. [33] are possible. Therefore, in the proposed RoleVR, spatial role is given to HMD user to maximize the sites' sense of space, and temporal role is given to Non-HMD user, wherein she/he can determine the overall situation through the flow of time and perform actions directly while leading conversations.

4.1.1 Spatial role of HMD user

The HMD user in charge of the spatial role receives the 1st-person view scene information as shown in Fig. 4b while wearing the HMD device as shown in Fig. 4a, and perceives the situation while experiencing directly in the virtual environment. In the proposed RoleVR application, the user must move safely to the end destination of the maze. One thing to note here is that the user must select all the check points to open the destination door, and they must never be touched by NPCs during this process. In addition, during this process, the user can perform additional actions (dropping bombs, etc.) to explore paths and collect items, that is, the user's role can broadly be divided into movement and actions.

RoleVR restricts the HMD user's core role to space. Therefore, HMD user focuses only on the movement role of exploring the maze, finding check points, and avoiding NPCs. Moreover, she/he can directly relay the current situation that she/he sees to Non-HMD user to share her/his presence (thoughts, situation). Figure 4 shows the process of performing the spatial role assigned to HMD user in the proposed RoleVR application. She/he is in charge of finding check points that are hidden in the maze and avoiding NPCs that block the user (Fig. 4c). HMD user directly feels the situations that occur on-site, so they require quick judgment and reflexes. This kind of role makes it possible for the basic HMD user to feel a high level of presence.

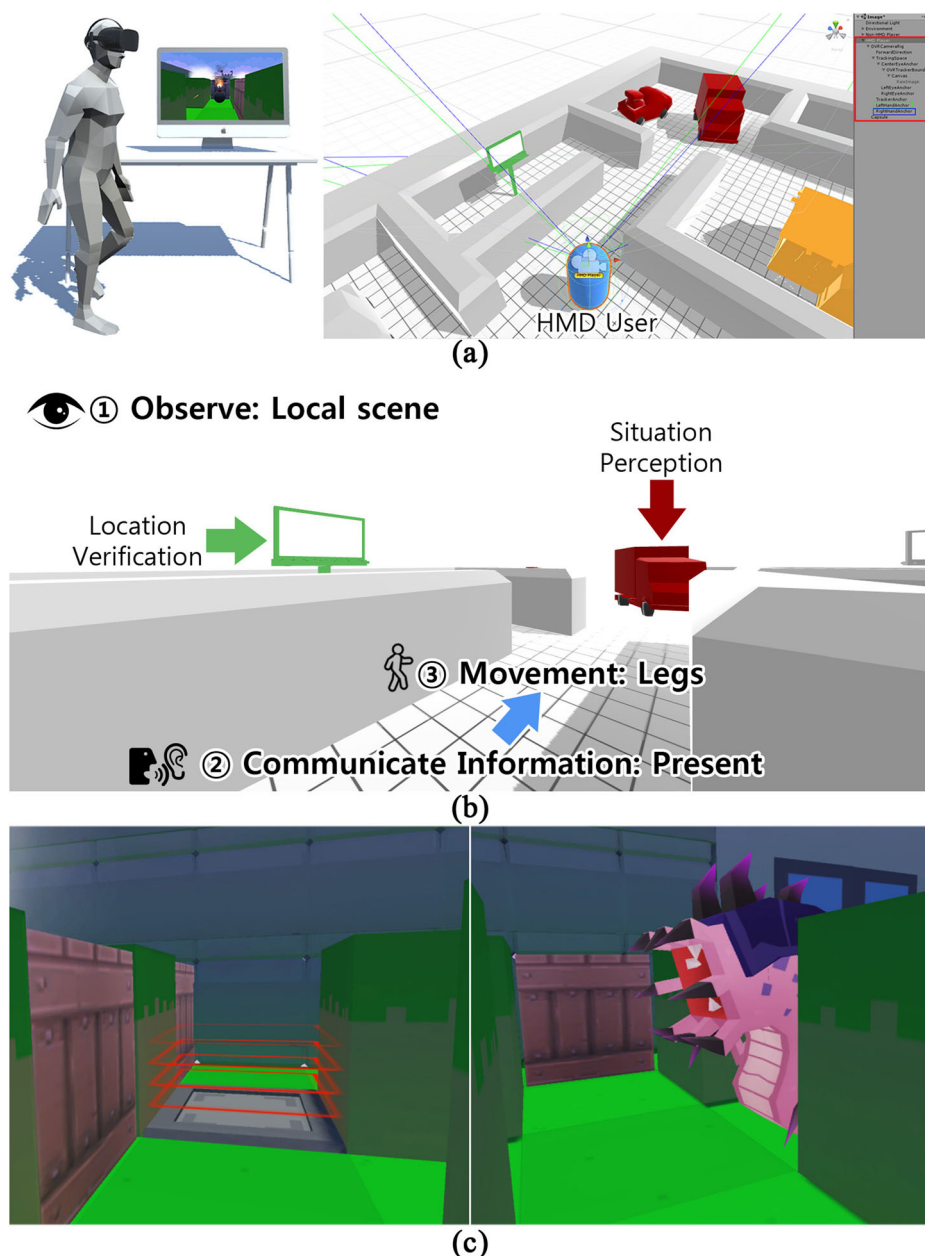


Fig. 4 Spatial role of HMD user: **a** configuration of spatial role, **b** generalized spatial role user's view, **c** spatial role of HMD user in the proposed RoleVR application (search of check point, NPC avoidance)

4.1.2 Temporal role of Non-HMD user

Next is the Non-HMD user's role. Due to the difference in the experience environment that occurs in IVR, it is unavoidable that Non-HMD user will feel a lower level of presence than

HMD user. In existing asymmetric VR games, there were limits to reducing the difference in presence due to problems related to the non-immersive systems' hardware. In Cheng et al. [4] and Gugenheimer et al. [9], the participation of Non-HMD users was expanded with actions but limited to an auxiliary role; therefore, it was unable to provide the same presence as it provided to HMD user.

In the proposed RoleVR, the Non-HMD user's role is approached through a time concept in order to expand it from leading the application through decisions on the overall situation to directly performing actions. If we define roles based on the RoleVR application, Non-HMD user is of course in charge of making overall situational decisions required to achieve goals, such as analyzing maze patterns and NPCs, seeing the locations of item boxes and check points, etc. The Non-HMD user relaying them to the HMD user, but Non-HMD user is also in charge of transferring these decisions into actions (dropping bombs, shields, etc.), which is the most important factor. Owing to this, Non-HMD user performs an equal or greater role than HMD user in the application. Figure 5 shows the Non-HMD user's temporal role. First, she/he analyzes the detailed factors needed to achieve their goals through the 3rd person camera view and make decisions about the overall execution approach. In addition, she/he directly performs a variety of actions along with the movement of HMD user as the application progresses.

4.2 Immersive interaction

Asymmetric interactions play an important part in communication between users in CVE. Immersive interactions are also a core factor in increasing the user's presence in an IVR. Therefore, in RoleVR, we have designed interactions that can increase presence by maximizing the immersion and communication based on the roles set for the two types of users. The core of the HMD user's spatial role in RoleVR is movement, and we designed a walking interaction using the user's two legs, while for the temporal role performed by the Non-HMD user, we designed a hand interface for increasing activeness and the realism of the action.

4.2.1 Walking interaction for HMD user

The first interaction is a walking interaction for the HMD user. The HMD user is in charge of the spatial role, and the core of that role is movement. Because movement uses the legs, it is important to design an interaction in which the legs can be directly used. In existing asymmetric VR games, movement is mainly done through a keyboard or gamepad. In the case of the research that uses space, expressing only the movement as the room scale motion provided in the HTC Vive is essential. However, the proposed RoleVR uses the walking simulator proposed by Lee et al. [20], which is portable, simple and easy-to-wear to express free walking.

It was confirmed that, when walking in place represents movement in the virtual environment, it provides vastly improved immersion and thus an increased presence compared to existing input processing techniques that use a gamepad or keyboard, regardless of the fact that there is no difference in cost or portability [20]. Therefore, in this study, we used an RF module to supplement delayed or missing data occurring during communication between the simulators of both legs based on the walking simulator proposed by Lee et al. [20], which determines walking-in-place of user through gradient changes of a gyro sensor (MPU6050) mounted on the simulator. Furthermore, we implemented an additional function for

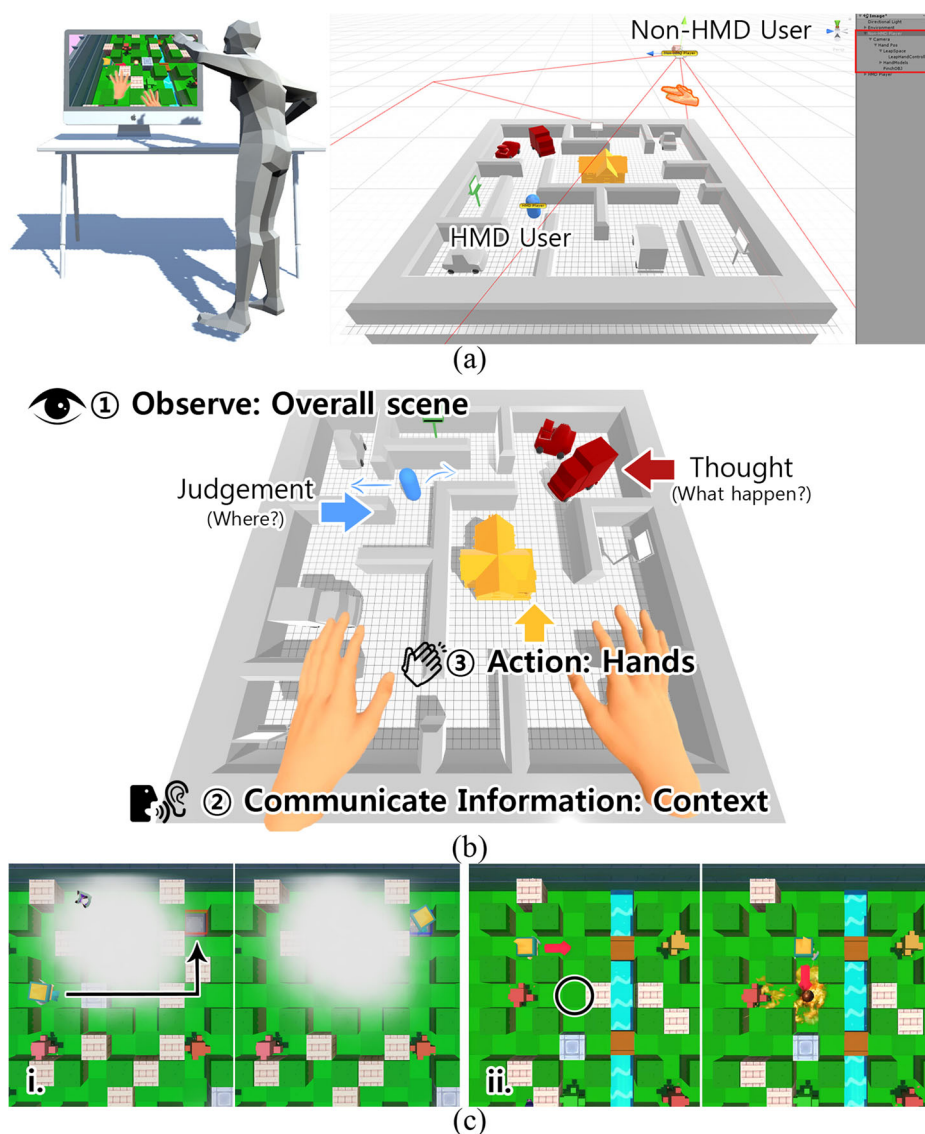


Fig. 5 Temporal role of Non-HMD user: **a** configuration of temporal role, **b** generalized temporal role user's view, **c** decision of overall situation (i. Non-HMD user detected unidentified area (thought) and inferred that a mysterious place might be dangerous (judgment), ii. Non-HMD user noticed that all roads are blocked (thought), thus figuring out the best path to go (judgment))

direction change, through the y- and z-axis acceleration gradient values of simulator gyro sensor mounted on the right leg.

Equation 1 is a process for determining the walking-in-place through the difference of acceleration gradient value detected by the gyro sensor of the walking simulator. It is determined as a view change (y-axis) or a walking state (z-axis) when the difference value (D) of the rotation axis of the gyro sensors is larger than the threshold range (T). At this time,

if the D value does not exceed the threshold range (all negative values), it is an stop state. Here, k is an index that computes changes in the y - and z -axis.

$$\arg \max_{k \in \{y, z\}} (D^k - T^k) \quad (1)$$

Figure 6 shows the walking simulator's system structure. While performing a spatial role, the HMD user wearing the walking simulator is provided with a high level of immersion in a simple and intuitive experience environment, and can control free movements.

4.2.2 Hand interface for non-HMD user

The hand is the most often used part of the body for capturing various user behaviors in an IVR. Therefore, while the HMD user's interaction focuses on the legs to consider movement, the Non-HMD user's interaction focuses on the hand. We designed a hand interface that can provide a realistic presence in the process of interacting with a virtual environment using a hand. It can be seen that the Leap Motion-based hand interface presented by previous studies [11] helps improve the presence through relatively high immersion in the IVR, compared to a simple control method using the gaze or a gamepad. Therefore, in RoleVR, we designed a Leap Motion-based hand interface for Non-HMD user that maximizes immersion in the process of interacting with the virtual environment with your hands and allows users to experience a high level of presence. The Leap Motion sensor handles hand motion capturing and gesture recognition, and on this basis, the proposed hand interface comprises a structure in which the menu and motions are predefined; a result event is executed when a corresponding user input is received.

Equation 2 determines the hand gesture of the current user through the difference between the pre-defined gestures (G) and the input gesture (I) using the finger bending (f), palm (p), and clapping (c) as parameters. Where H is the set of defined hand gestures, x is the index to search by the defined gesture. The gesture information (G) is set in advance by H .

$$\arg \min_{x \in H} \left(\sum_{i=1}^5 \sqrt{(G_f^x(i) - I_f(i))^2 + |G_p^x - I_p| + |G_c^x - I_c|} \right) \quad (2)$$

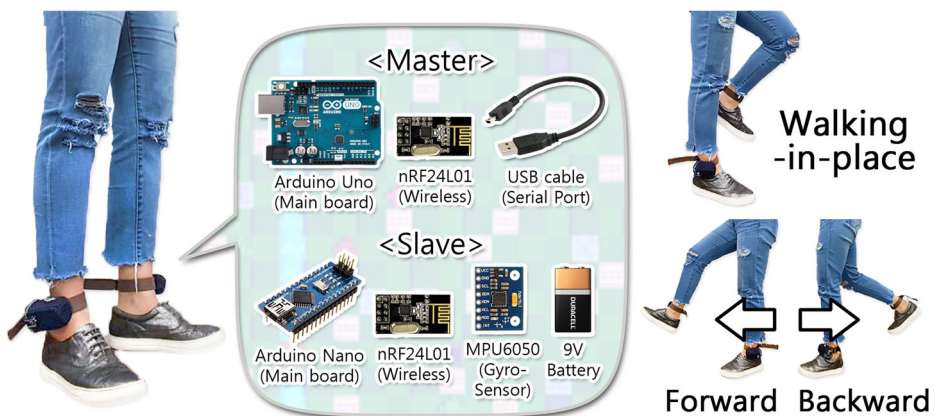


Fig. 6 System structure of walking simulator based on walking-in-place used by the proposed RoleVR

Figure 7 shows the process of controlling the application through the Non-HMD user's hand interface. In the proposed RoleVR application, the Non-HMD user performs 5 kinds of actions through the predefined gestures.

4.3 Understanding of roles

This section explains how to improve the understanding of roles based on cases that present concrete scenarios. It is important for each Non-HMD and HMD user to perform their roles correctly, because the role within the proposed RoleVR is shared by all users in one virtual character. Otherwise, it may cause a disparity in the level of presence between HMD and Non-HMD users. It may alternatively present a monotonous experience. This study confirms scenarios in the proposed asymmetric VR application for identifying the importance of the role of each user. Figure 8 shows examples. To achieve a goal in a given situation, all users together must perform a mission. Each mission has solutions that are pre-encoded. Here, a Non-HMD user identifies solutions quickly and conducts his own role by communicating with HMD users. However, the solution is just one example that is presented. Thus, the user may find and perform other solutions. Note that the temporal role of the Non-HMD user, who can judge the situation as a whole, is important. Conversely, if the HMD

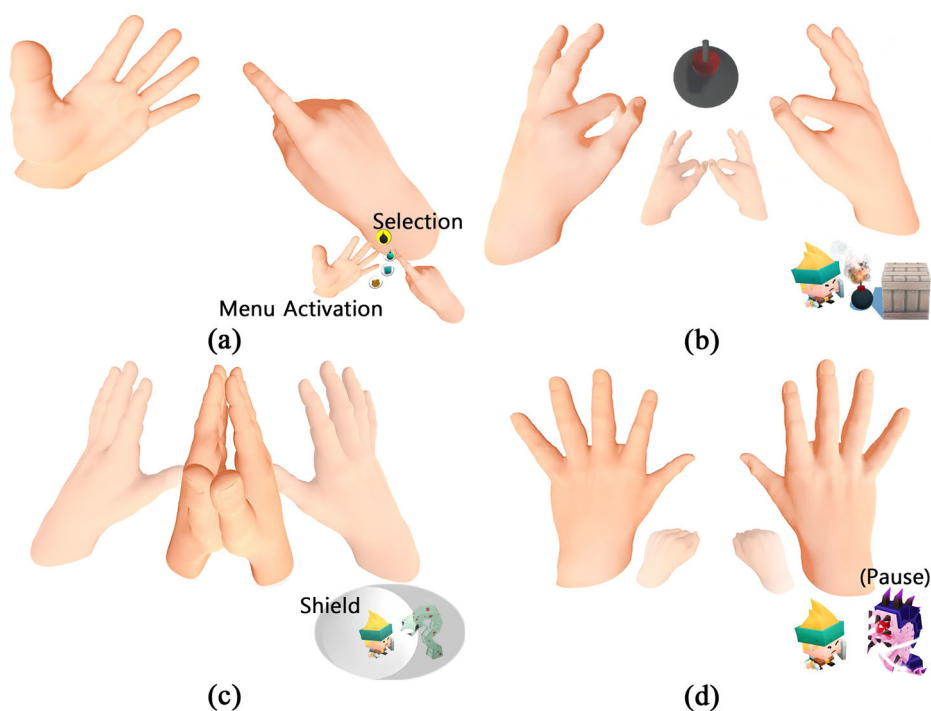


Fig. 7 The proposed hand interface for a Non-HMD user (parameters left($f[]$, p , c) & right($f[]$, p , c), action): **a** open one's palm and click with a fingertip ($([0,0,0,0,0],1,0)$ & $([0,1,0,0,0],0,0)$, menu activation and menu selection); **b** with both hands pinch and release ($([1,1,0,0,0],0,1 \rightarrow 0)$ & $([1,1,0,0,0],0,1 \rightarrow 0)$, bomb-dropping); **c** clap one's hands ($([0,0,0,0,0],0,0 \rightarrow 1)$ & $([0,0,0,0,0],0,0 \rightarrow 1)$, shield activation); **d** open one's hands ($([1,1,1,1,1] \rightarrow [0,0,0,0,0],0,0)$ & $([1,1,1,1,1] \rightarrow [0,0,0,0,0],0,0)$, time control)

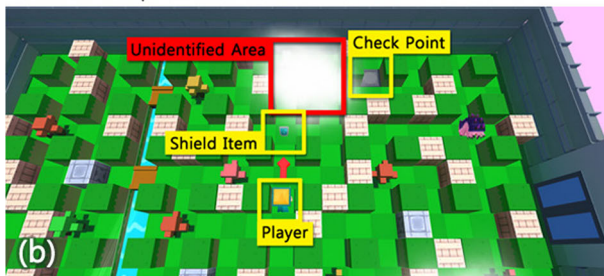


Mission : go to bottom check point

Solution.

Non-HMD : path information delivery to HMD after shortest path detection and destroy box

HMD : move path based on delivered situation



Mission : move to check point after item acquisition

Solution 1.

Non-HMD : use shield item and send check point location

HMD : move quickly after cofirming use of shield item

Solution 2.

Non-HMD : identify bypass path and destroy box

HMD : move accurately path based on delivered situation



Mission : escape to destination avoiding NPC

Solution 1.

Non-HMD : use selected item(shield, timer, etc) after sharing strategy

HMD : move quickly after cofirming use of selected item

Solution 2.

Non-HMD : send accurate starting time after timing calculation

HMD : move quickly by avoding NPC

Fig. 8 Examples of understanding the roles of the Non-HMD user and the HMD user, depending on the specific situation

user with limited visual information leads the mission, it becomes difficult to carry out. Consequently, confusion can increase and interest can be reduced. Additionally, Non-HMD

users may cause problems, because a disparity in the level of presence may occur due to a narrower activity scope.

In Fig. 8a, if a Non-HMD user guides an HMD user in a way that can be bypassed and destroys the boxes blocking a road, the virtual character can easily move to the check point. However, if a Non-HMD user follows the HMD user's movements without any feedback, the Non-HMD user is placed into a passive role, looking at the HMD user's behaviors to avoid the NPC. Thus, the HMD user probably will not avoid the NPC. In the end, it can be confusing if the user cannot escape the virtual space. As shown in Fig. 8b and c, all users can immerse themselves in the situation when a Non-HMD user sets the strategy, uses the appropriate item, and moves the HMD user around quickly. However, when the Non-HMD user takes a passive role, the HMD user suffers fatigue because of unnecessary efforts. Thus, the Non-HMD user fails to do more work. Consequently, participation and interest rapidly decreases. Based on these scenarios, diverse experiments are conducted to confirm the effects of the proposed role on the user's presence and experience (Section 6).

5 Environments

The RoleVR's application and the co-located users' experience environment were implemented using Unity 3D 5.3.4f1, Oculus SDK (ovr_unity_utilities 1.3.2), and Leap Motion SDK v.4.1.4. Furthermore, the walking simulator used to control the HMD user's movement is based on the work of Lee et al. [20] and was newly designed using Arduino sketch v.1.6.12. Finally, the PC used to build and test the integrated development environment was equipped with an Intel Core i7-6700, 16GB RAM, Geforce 1080 GPU.

Figure 9 shows the application built to analyze the presence and multi-experience between co-located HMD and Non-HMD users provided by the proposed RoleVR. It shows the entire process of performing actions that the main character makes, from leaving the starting point, avoiding NPCs, and collecting check points/items to arriving safely at the goal.

Next is the structure of the environment considering each interaction and communication between co-located HMD and Non-HMD users. Two users can enjoy the experience together in a space with a width of 2.5m and a length of 2.5m. Monitors are installed so that the Non-HMD user can see the 3rd person's view scene that shows the overall situation, as well as a scene that shows the 1st person view of the HMD user, which is meant to

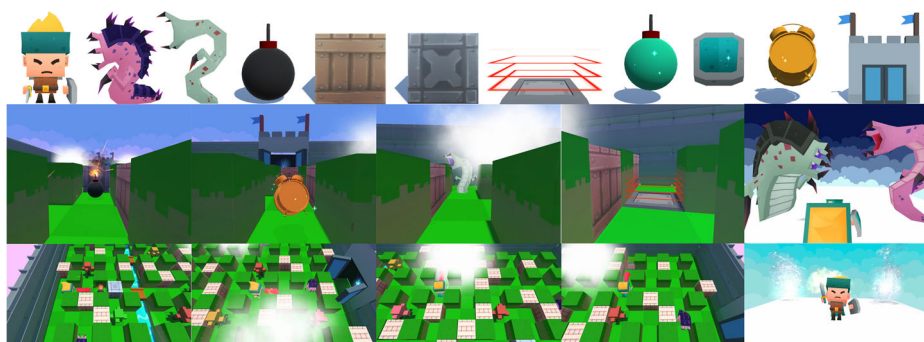


Fig. 9 Results of the proposed asymmetric VR application (top: objects used in the proposed asymmetric VR application, middle: HMD user view, bottom: Non-HMD user view)

help improve the presence by allowing the Non-HMD user to indirectly experience what the HMD user is experiencing. Furthermore, Leap Motion is installed in front of the 3rd person view monitor for the Non-HMD user's interactions so they can take control of the overall situation. Figure 10 shows the RoleVR experience environment.

6 Experimental results and analysis

6.1 Experiments and participants

The core of the proposed RoleVR is to have the co-located users who are experiencing the asymmetric VR application feel a similar level of presence while experiencing overall satisfaction (enjoyment and social interaction) with the application and having their own individual experiences based on their roles. To systematically analyze the purposes of RoleVR, we performed a survey based experiment. The survey participants consisted of 20 person between 20 and 35 years old who had experience with single-player VR contents, and whose average age was 25.8 years. A relatively young age group who could quickly respond to dynamically changing situations and actively communicate with others was chosen. The proposed RoleVR is played in pairs of one HMD and one Non-HMD user. Therefore, the participants were divided into 10 paired teams for this experiment.

6.2 Overview of experiments

The survey experiment comprises the presence, experiences (social interaction, enjoyment), and suitability of immersive interaction of co-located HMD and Non-HMD users in asymmetric virtual reality. First, it is systematically analyzed whether there is a change in presence feels like "being there together" through independent roles based on the presence of HMD users and the sharing of experiences with Non-HMD users. Here, presence

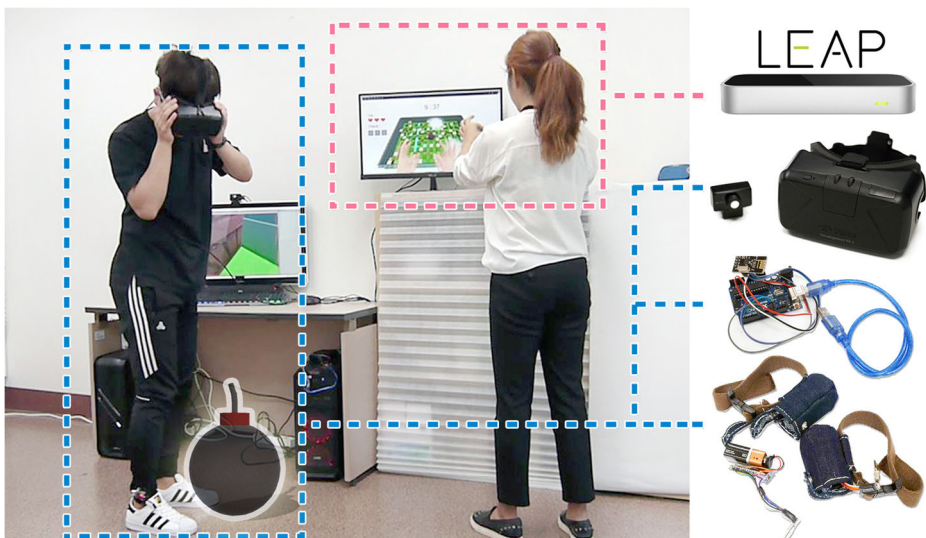


Fig. 10 Construction of the RoleVR co-located experience environment

is collected through reliable survey data by utilizing Witmer et al.'s presence questionnaire [39]. The presence questionnaire was a questionnaire with reliability (i.e., Cronbach's alpha) of 84% or more. It comprise a total of 19 questions, including "How natural did your interactions with the environment seem?", "How much delay did you experience between your actions and expected outcomes?", among others. The user selects and records a value between 1 (not at all) and 7 (completely) for each question. It is composed in such a way that it divides and analyzes five detailed factors such as realism, quality of interface along with comprehensive decision about presence.

Furthermore, the satisfaction and difference of the experience composed of social interaction and enjoyment are analyzed in the same manner. Moreover, whether each user has an accurate understanding of his/her role is checked. For the survey on social interaction and enjoyment, this study utilizes the game experience questionnaire (GEQ) [13]. The survey participants record a value between 0 (not at all) and 4 (extremely) for the questions presented in the GEQ. The survey on social interaction comprises 17 questions. Enjoyment can be analyzed through the questions 1, 5, 7, 8, 12 and 16 ("I felt revived", "I felt satisfied", among others) among the post-game module that is composed of the 17 questions ("I empathized with the other(s)", among others). Law et al. [19] confirmed that the survey questions for GEQ had reliability of 65% or more (Cronbach's alpha). Finally, to improve the immersion in the role, whether the immersive interaction applied to HMD and Non-HMD users is suitable for the asymmetric experience environment of RoleVR is analyzed and verified based on the survey results.

6.3 Experimental method and strategy

The experimental design of this study basically used a specialized questionnaire on presence and experience to increase the reliability of the results. Furthermore, considering the effects of the experience sequence on the survey results, 10 of the 20 participants first played the HMD role and then played the Non-HMD role, and the other 10 participants first played the Non-HMD and then the HMD role. One important point here is that, to analyze accurately the relationship between the presence ("being there together") and role provided through experience sharing based on the sense of space of HMD users from the viewpoint of Non-HMD users, the role of the Non-HMD users was divided into three steps, and the participants experienced them separately. In other words, each participant has four experiences (HMD user and three steps of Non-HMD user). As the interaction between HMD and Non-HMD users in the traditional asymmetric VR is optimized for the proposed experience environment and system, conducting direct comparative experiments with this study is impossible. Therefore, this study conducts experiments by dividing them into the three steps of the experience environment in order to maximally consider and comparatively analyze the roles of Non-HMD users in the traditional asymmetric VR.

The first of the three-step roles of the Non-HMD user is the audience role, in which a subject sits and observes the third-person screen and the first-person view of the HMD user. The second is an assistant role, in which a one-way passive participant grasps the information generated during the progress of the application through the third-person screen and communicates it to the HMD user. The last step is a temporal role of the proposed RoleVR, which is a participant with an interactive independent role.

In this experimental environment, comprising both HMD and Non-HMD, the role and immersive interaction of each user based on the experience environment are defined as independent variables, and the presence and experience through them are defined as dependent variables [23].

6.4 Presence

6.4.1 Analysis of presence according to role

In the first experiment, we analyzed the responses from the questionnaire that compared the co-located users' presence. In RoleVR, the aim was to have all users in the CVE feel a similar level of presence regardless of whether they were equipped with the HMD or not. In this study, we used the presence questionnaire proposed by Witmer et al. [39] to gather specialized presence survey data. In Fig. 11, it can be seen that the difference in presence between HMD and Non-HMD users using the RoleVR system was not large (M: 5.85, 5.72, SD: 0.66, 0.70). The proposed RoleVR gives the user the optimal role (spatial and temporal) based on whether a user is wearing the HMD or not. Furthermore, we designed an experience environment, in which each user can focus on his/her role and collaborate with each other, along with immersive interaction that supports it. As a result, regardless of whether the HMD is worn or not, improved overall presence can be obtained.

A detailed analysis showed that the difference of presence amplifies when the Non-HMD user is in-charge of the temporal role, who is relatively less active, in a passive stance. Because the HMD user can be active in the virtual space on the basis of three-dimensional visual information, experiencing basic presence is assured; thus, it is confirmed that the attitude of a Non-HMD user dealing with experience induces a difference in presence. By contrast, if the Non-HMD user is in an active stance, the HMD user can be commanded as an avatar, and in this process, he/she is assimilated with the movements and actions of the HMD user, thereby immersing into a situation as if experiencing both roles. In this case, the presence felt is higher than that of the HMD user. Thus, we confirm that all users can be provided with high level of presence when optimized roles are given to the users in the asymmetric experience environment and the conditions are satisfied whereby each user (especially non-immersive user) can participate in the application actively. The calculation results of statistical significance for presence through one-way ANOVA analysis confirmed

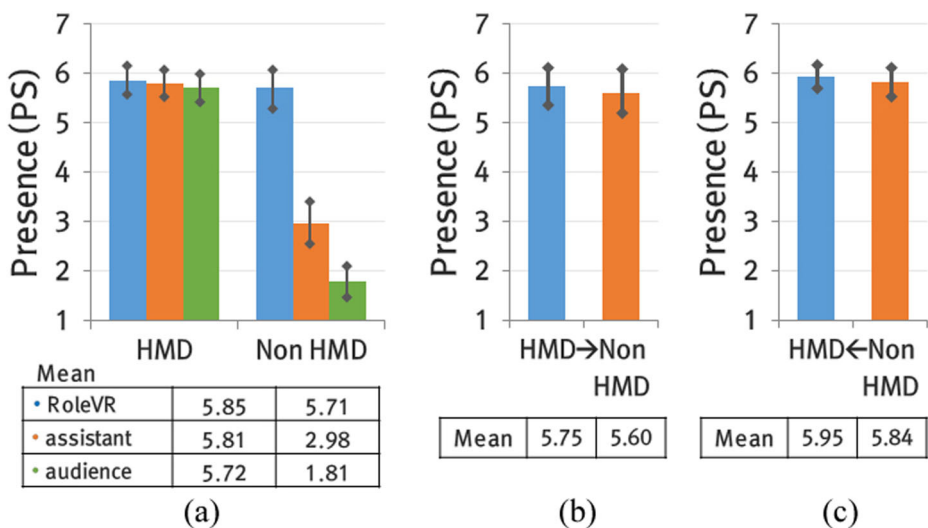


Fig. 11 Presence analysis results between co-located HMD and Non-HMD users: **a** presence comparison, **b** experience order: HMD → Non-HMD, **c** experience order: Non-HMD → HMD

that there was a significant difference in presence between the Non-HMD and HMD users when the Non-HMD users play a role as assistants or audiences. However, the roles and interaction of Non-HMD users in the proposed RoleVR were not statistically significantly different from those of the HMD users (Table 1).

Additionally, as we assumed that the order in which the roles were performed can cause a difference in presence. The results were divided based on the order in which the HMD was worn and the Non-HMD role was performed. This is seen in Fig. 11b and c. In the case of HMD users, understanding the proposed asymmetric application’s overall situation through the temporal role and then having the experience improved presence, and for the Non-HMD user, a high level of presence was shown when the temporal role was experienced first. The difference in the effects of experience order on the difference of presence was due to the Non-HMD user’s temporal role characteristics. Similar to a human brain, temporal role starts with an overall understanding of the application. Therefore, the presence improved when wearing the HMD after understanding the goal, action, and direction provided by the application. As a result, this study attempted to give increased priority to the temporal role of Non-HMD users, and it was confirmed that the temporal role directly affects the presence. To analyze presence separately for the HMD user and Non-HMD user roles based on the detail items of the presence questionnaire, the role of HMD users showed the highest realism due to the stereoscopic visual information in the virtual environment, as expected. In contrast, the role of Non-HMD users recorded higher values in possibility to act and quality of interface, which are related to making judgment about the situation and operating the system in a virtual environment. In other words, although Non-HMD users experience lower presence in space compared with HMD users, they have direct influence on the behaviors of HMD users and the overall situation, and, as a result, they experience the resulting changed environment together with HMD users. This confirms that advanced presence (“being there together”) is provided in the asymmetric virtual reality that is pursued in this study.

6.4.2 Relationship between the temporal role and presence of non-HMD users

In this study, to specifically study the effect of temporal role on the presence, we conducted comparative experiments on the role of Non-HMD, by dividing it into three steps. Step 1 comprises a regular VR application structure, in which the Non-HMD user (audience role) observes, whereas the HMD user performs all roles. Furthermore, in an existing asymmetric VR like Share VR [9], the assistance role performed mainly by Non-HMD is the step 2 experiment environment. Here, the Non-HMD user (assistant role) provides only the information (location, surrounding situation, etc.) required for processing with the application, and the HMD user performs all actions directly. Furthermore, step 3 is the experiment environment of the proposed RoleVR. The comparison graph of Fig. 11a confirms that, similar

Table 1 Comparative analysis result of presence between the HMD users and Non-HMD users

Pairwise comparison	
RoleVR: HMD vs. Non-HMD	$F(1,38)=0.372, p=0.545$
assistant: HMD vs. Non-HMD	$F(1,38)=359.740, p<0.001^*$
audience: HMD vs. Non-HMD	$F(1,38)=661.618, p<0.001^*$

* mark indicates statistical significance

to the presence of existing asymmetric VR, steps 1 and 2 show a clear difference of presence from the HMD user (M: 5.71, 2.98 and 1.81). Thus, in the asymmetric environment, assigning a role to the user is more influential than wearing an immersive device like HMD.

6.5 Experiences in asymmetric VR

6.5.1 Social interaction and enjoyment according to role

In the next experiment, we systematically analyzed the experience of providing roles, which are the core factors in RoleVR. To do this, we used the GEQ [13]. We evaluated the users' satisfaction with RoleVR's social interaction through the psychological involvement (empathy) items and behavioral involvement items in the social presence module of the questionnaire. We evaluated the users' satisfaction with enjoyment through the positive experience items in the post-game module of the questionnaire. Figure 12a and b show the results of HMD users and Non-HMD users, respectively. It can be seen that Non-HMD users have relatively high values for both factors, while at the same time all users experienced satisfaction with little difference (social interaction M: 3.63, 3.73, SD: 0.25, 0.23, enjoyment M: 3.30, 3.42, SD: 0.39, 0.51).

It is worth noting that interest and social interaction are both highly satisfying for Non-HMD users. As described above, given that the Non-HMD user gives much meaning to the temporal role (situation judgment, conversation) in terms of controlling and experiencing the application, it is possible to directly experience the process of predicting and determining sudden situations or the entire situation so that further immersed could be enabled. As the HMD user performs instantaneous movements based on the information provided by the Non-HMD user, experience on the space is high, but she/he is relatively passive in the process of focusing on the application and consequently the score recorded was low. However, this also caused HMD users to become more interested when the HMD user is in charge of the situation as a whole or when the proposed immersive interaction meets the role appropriately. Lastly, Table 2 shows the calculation results of one-way ANOVA for

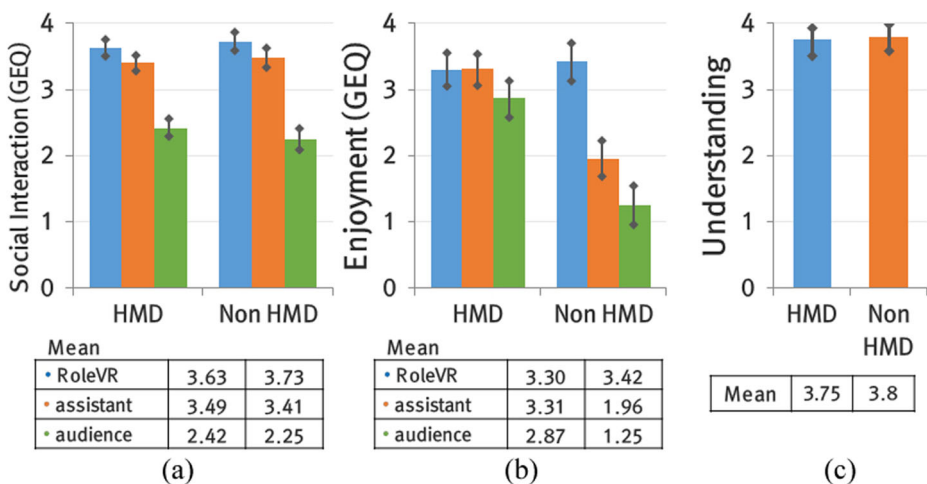


Fig. 12 Experience analysis results between HMD and Non-HMD users: **a** social interaction, **b** enjoyment, and **c** understanding for each role

Table 2 Comparative analysis result of social interaction and enjoyment between the HMD users and Non-HMD users

Pairwise comparison	
Social Interaction	
RoleVR: HMD vs. Non-HMD	$F(1,38)=1.766, p=0.192$
assistant: HMD vs. Non-HMD	$F(1,38)=1.011, p=0.321$
audience: HMD vs. Non-HMD	$F(1,38)=5.898, p<0.05^*$
Enjoyment	
RoleVR: HMD vs. Non-HMD	$F(1,38)=0.629, p=0.433$
assistant: HMD vs. Non-HMD	$F(1,38)=107.773, p<0.001^*$
audience: HMD vs. Non-HMD	$F(1,38)=264.388, p<0.001^*$

* mark indicates statistical significance

social interaction and enjoyment. It indicates that there is no statistically significant difference in the roles and interaction of all users in the proposed RoleVR. However, when the Non-HMD users play the role of assistants, they are statistically shown to be within similar ranges due to the experience environment where the HMD user can talk directly with the HMD users in the same space (co-locate). In all other cases besides this, the Non-HMD users have a statistically significant difference to the HMD users.

6.5.2 Analysis of user's understanding about role

Here, it is necessary to confirm that each user accurately understood the role provided by RoleVR and that they actually experienced the role. To do this, we divided the responses regarding the experience via roles and assigned values to them, with 0 meaning strongly disagree and 4 meaning strongly agree. We gave HMD users the statement “I was in charge of a spatial role, and I fully received a spatial experience”, and we gave Non-HMD users the statement, “I was in charge of a temporal role, and I commanded the overall situation and led the action”. Figure 12c shows the values recorded for this, and it can be seen that both kinds of users accurately understood their roles and had a satisfactory experience suitable for the situation (M: 3.75, 3.8, SD: 0.43, 0.40). The results of this survey show that most of the users who participated in the experiment correctly understood the role of RoleVR as proposed in this study and also learned how to control the immersive interaction.

6.5.3 Relationship between the temporal role and experience of non-HMD users

For the second experiment, we conducted a comparative experiment by dividing it into three steps (Fig. 12). In step 1, as we minimized the role of the Non-HMD user (audience role), low experience values were provided in all aspects. In step 2, as the information provided by the Non-HMD user (assistant role) led to direct response and action of the HMD user, the social interaction was marginally improved (M: 3.73, 3.41 and 2.25). However, with regard to enjoyment, the Non-HMD user could not overcome the limitation of assistant (M: 3.42, 1.96 and 1.25). Furthermore, with regard to experience, the direct influence of a role was large in the asymmetric VR environment. In this study, however, **since** more weight was given to the temporal role of the Non-HMD user, the presence and satisfaction of experience was further increased. Therefore, immersion and experience provided satisfied

all users when the role of the Non-HMD user was defined to suit the goal and structure of an asymmetric VR application.

6.5.4 Suitability of the proposed immersive interaction

Finally, we analyzed whether the immersive interaction of this study facilitated performing the role. We were able to confirm this by comparing the “possibility to act” values from items 1, 2, 8 and 9 and the “possibility to examine” values from items 15 and 16 in the result data obtained from the previous presence questionnaire. In the results, the “possibility to act” had a mean of 5.86 (SD: 0.69) for HMD users and a mean of 5.94 (SD: 0.88) for Non-HMD users, and the “possibility to examine” had a mean of 6.23 (SD: 0.48) for HMD users and 5.72 (SD: 0.77) for Non-HMD users, so we were able to confirm that the presence was helped by an improvement in the role’s immersion owing to both proposed interactions (walking simulator and hand interface). Because we confirmed that the proposed interaction (walking simulator and hand interface) provided better presence than the base line (gaze-pointer, keyboard and gamepad) through the previous studies [11, 20], we did not perform the comparison; however, we found that the proposed interaction by itself provided enough positive help.

7 Discussion

In this study, to specifically analyze the effect of a user’s role on the presence and experience in the asymmetric VR experience environment, we performed confirmation and verification by providing a single application as an example. This analysis can be interpreted as a method applicable to only certain applications or limited experience environment. However, we can use it in any area under the generalized concept that the Non-HMD user has the temporal role of understanding and determining the situation overall, and the HMD-user has the spatial role of handling movements in the space. Figure 13 shows examples of applying it to various situations under generalized roles, and a virtual house whereby we applied the temporal role to the Non-HMD user as a guide delivering the information inside the house and the spatial role to the HMD user as a consumer experiencing the house (Fig. 13a). In Fig. 13b, we constructed a VR farm casually and the Non-HMD and HMD users are classified as supervisor and worker, respectively, and their roles are applied accordingly. Similarly, when we distinguished the proposed roles in an asymmetric VR to meet the goal of the application, it was applicable in various areas. Furthermore, we believe that all users are provided with satisfactory presence and experience during the experience process, as confirmed through our experiments.

Furthermore, independent immersive interaction was proposed to improve the immersion of HMD and Non-HMD users in their roles, so that every user can experience improved presence. Furthermore, it was verified through previous research that the interactions (walking simulator, hand interface) optimized for each user provide improved presence and satisfactory experience through immersion compared with general input devices (e.g., keyboard, mouse, game controller, or gaze pointer). Lee et al. [21] demonstrated that controlling the motions of virtual characters using walking-in-place have the effect of not only greatly improving presence compared with game controller and keyboard, but also reducing VR sickness that can happen in virtual reality. It has been also confirmed through various comparative surveys that the hand interface for controlling virtual environments or objects using the hands of users produces more-satisfactory results than gaze pointers frequently used in

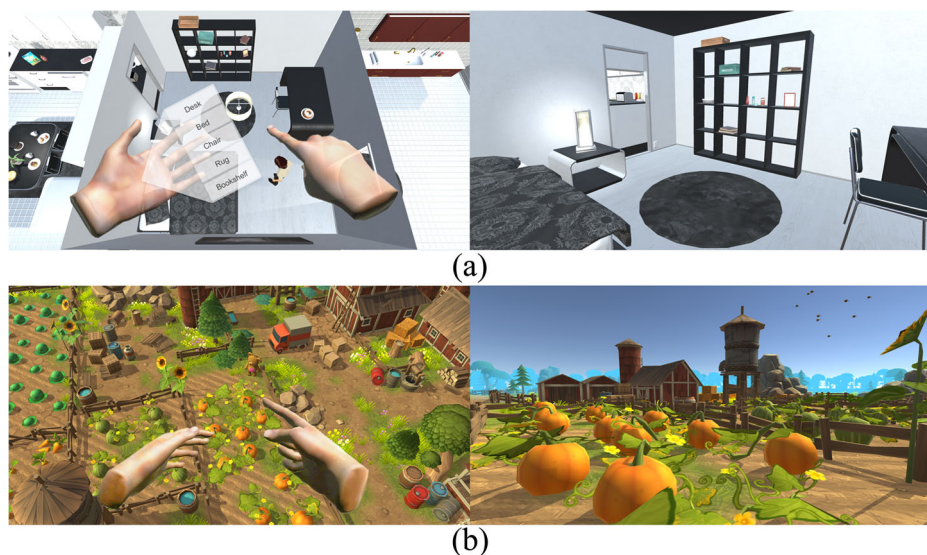


Fig. 13 Examples of the application direction of the proposed RoleVR (Non-HMD and HMD user's viewpoints): **a** VR house (interior), **b** VR farm

VR applications and also has the advantage of reducing VR sickness [11]. Moreover, it was developed in an application study that proposed a new VR interaction method of the third-person viewpoint using this hand interface [16]. In addition, various motion platforms (e.g., treadmills) and haptic interfaces are being researched. Therefore, one should consider comparison studies with various systems and interaction methods other than the baseline and directions for applying them to interactions in asymmetric VR.

8 Conclusion

In this research, the main purposes of RoleVR are to provide co-located HMD and Non-HMD users with a variety of experiences and to have both types of users feel the presence together in an IVR, and to provide an environment that can be satisfying. To do this, RoleVR discerns the difference in the users' systemic and experiential environments and suitably differentiates roles for the users. Here, RoleVR places more focus on the Non-HMD user. The HMD user is assigned to a spatial role that focuses on movement, and the Non-HMD user is assigned to a temporal role in which she/he is in charge of actions based on judgments and decisions about the overall situation. We also designed interactions (spatial role $\leftrightarrow$ walking simulator, temporal role $\leftrightarrow$ hand interface) that can maximize the immersion of the roles based on systems and interfaces that have been verified through previous work. As a result, the interactions that underpin the roles make it possible to improve the presence of all users in the virtual environment. We created an "asymmetric VR application" that emphasizes the communication and cooperation between co-located users. The application was confirmed by surveys [9], and it features interactions that are optimized for clearly differentiated roles. This was used in experiments to verify the proposed RoleVR system. Finally, general participants ($n = 20$) experienced the created asymmetric VR application in HMD and Non-HMD environments, and the results were recorded. The results were

able to confirm that all of the users who the RoleVR targeted could feel a similar level of presence in the virtual environment. Moreover, because of this, the majority of users responded by saying that the roles played an important part. The survey responses about the experience revealed that all users felt high and similar levels of enjoyment and social interaction, and even more importantly, they understood the roles that they were given in the user environment and felt experiences suitable to their roles. Based on these results, we have proven that roles, including interactions, can be used as an important factor in making Non-HMD user, who are placed in a relatively non-immersive environment, as immersed as HMD user in the IVR.

In this study, we divided the roles of HMD user and Non-HMD user into spatial role and temporal role, respectively. However, we checked the spatial role and temporal role cases only through a single application. In future work, we will create applications that employ a variety of concepts, genres, etc. in a competitive structure rather than a cooperative structure to expand the range of these roles and generalize them. In addition, we will perform survey experiments on a large number of participants.

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